

Challenges and Opportunities in Developing Systems to Assist Patients in Recovering from Gambling Addiction

Min-Wei Hung
National Tsing Hua University
mwhung@mx.nthu.edu.tw

Chi-Ting Hou
National Tsing Hua University
dafdafdaf0411@gapp.nthu.edu.tw

Chieh-Jui Ho
National Tsing Hua University
jerryjp3914@gapp.nthu.edu.tw

Hsueh-Sung Lu
National Tsing Hua University
leon19970102@gapp.nthu.edu.tw

Chien Wen (Tina) Yuan
National Taiwan Normal University
tinayuan@ntnu.edu.tw

Nanyi Bi
National Taiwan University
nanyi.bi@gmail.com

Shu-Wei Liu
Taipei City Hospital Songde Branch
hitdre@gmail.com

Ming-Chyi Huang
Taipei City Hospital Songde Branch
mch@tpech.gov.tw

Chuang-Wen You
National Tsing Hua University
cwyou@mx.nthu.edu.tw

ABSTRACT

Gambling addiction is marked by compulsive gambling behaviors despite negative consequences in patients' financial and social well-being. We conducted a needs assessment study and identified current difficulties to self-monitor gambling behaviors using existing technology solutions, as well as the gap to incorporate family supports in the system design. We then proposed a preliminary system design to make the information sharing among patients and family members possible by designing easy-to-use interfaces for self-reporting that allow patients' positive interpretations, facilitating micro-sharing for family members to bring in early supports, and incorporating treatment teams knowledge of gambling addiction to interpret collected data.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**.

KEYWORDS

Gambling addiction; Self-monitoring; Family support

ACM Reference Format:

Min-Wei Hung, Chi-Ting Hou, Chieh-Jui Ho, Hsueh-Sung Lu, Chien Wen (Tina) Yuan, Nanyi Bi, Shu-Wei Liu, Ming-Chyi Huang, and Chuang-Wen You. 2021. Challenges and Opportunities in Developing Systems to Assist Patients in Recovering from Gambling Addiction. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3460418.3479298>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA
© 2021 Copyright held by the owner/author(s).
ACM ISBN 978-1-4503-8461-2/21/09.
<https://doi.org/10.1145/3460418.3479298>

1 INTRODUCTION

Gambling addiction has been one of the important issues of global public health concern. Studies have shown that the increased popularity of internet gambling is resulted from the high accessibility, engaging setting, and the ease to spend money [5]. In spite of their desperation to quit, it is difficult for the gamblers to regulate behaviors on their own. To assist their recovery, it is important to involve professional support as well as family support to provide critical aids with active observations and timely interventions in daily life. Governments and researchers are putting the effort in developing mobile and PC applications to battle the gambling disorder [1, 4]. Specifically, qualitative studies were conducted to explore the role of persuasive technology and partner's influence in regulating gambling behaviors [3]. To our knowledge, little has been done to integrate the support of self-monitoring data, family members, and the treatment team altogether for gambling addiction. Built on existing works, this study presented findings from interviews with patients, treatment experts and family members, and proposed a phone-based supporting system to effectively monitor gambling behaviors and daily routine of patients with gambling addiction, and enable treatment teams and family members to provide support through data-sharing.

2 NEEDS ASSESSMENT STUDY

We conducted semi-structured interviews with participants to understand their experience of managing gambling-related problems and extract insights for designing assistive systems. We recruited four gambling-addicted patients (P01 ~ P04) aged 26-31 years, and one family member (F01) aged 63 years. For treatment experts, we interviewed a psychiatrist and a psychotherapist (TE01 and TE02). We audio-recorded and transcribed their responses to perform thematic analysis [2]. The findings are presented as follows.

2.1 The Difficulties of Self-monitoring Gambling Behaviors

For gambling addiction, treatment experts would ask patients to manually self-report their impulsivity, gambling cognition, and behavioral adherence, which can be used to assess or evaluate patients' progress. TE01 commented "These patients usually do not

self-monitor their gambling behaviors. We would ask them to manually record the gambling expenses, the number of gambling ... These are important indicators to evaluate treatment efficacy.” Moreover, gambling addiction may cause disruptive daily routine, considering the time they deprive of sleeping for gambling and the stress that comes with it. Various apps were designed to provide individuals with self-monitoring or self-reporting tools [1]. However, these technology solutions did not collect those lifestyle factors to reflect the impact of problematic gambling, making it hard for patients to beware of their gambling problems in daily life.

In addition, those who gambling online can place a bet anytime and anywhere through their mobile phone easily. P04 commented “I would place bets in between other daily activities. I do not need to switch my focus to that (betting). It only takes 10 seconds to place a bet.” Although existing self-monitoring technology allows patients to self-report the gambling-related data, most participants expressed that they don’t understand how self-report can help reduce gambling. P03 commented “[How would you interpret the collected gambling-related data?] The data would be meaningful if I have a benchmark to compare with. I mean, teach me how to read the data. Otherwise, I don’t know what I can do with just my data.” Some participants were afraid that the data they self-reported (or automatically logged by websites) might trigger their irrational beliefs, prompting them to keep chasing luck. P02 commented “[How about reading these data afterward?] I guess it would produce the opposite effect. The data would remind me that I have lost so much money that I need to turn it around by gambling again.” To complement with self-monitoring technology solutions, all participants expressed that timely family intervention did prevent them from gambling. Therefore, identifying appropriate ways to incorporate family support through technology should be explored further.

2.2 The Gap When Designing Technology to Incorporate Family Support

Due to heavy debt caused by patients’ problematic gambling, family members usually had conflicts or arguments with patients and felt exhausted to deal with patients’ financial burden. Despite their poor communication, the family participant expressed that he was still eager to help his child to cope with gambling addiction. However, the family participant expressed that currently he faced a problem of not having enough data to understand his son’s gambling behaviors. F01 commented “Honestly, I am not sure if he continues to gamble or how he bets. [Do you know if he stopped gambling for a month?] I am boldly assuming that he knows better (not to gamble). [...] However, it is really hard to say how much time he spent on gambling because he placed bets with his phone (just like normal phone usage).”

Nevertheless, all patient participants expressed concerns that their family members did not understand the nature of gambling addiction as a mental illness. Given this situation, family members might use the data collected by technology solutions as evidence of gambling but not as a reminder to introduce their supports. P03 commented “Family members need to be guided, perhaps by treatment team, on how to communicate with patients regarding the shared-data. Otherwise, they would only notice the negativity and blame patients, who would feel worse and resist using this system.” Given these observations, designing technology to bridge the gap

between the patients and family members requires more research to accommodate concerns raised by both sides.

3 PRELIMINARY SYSTEM DESIGN

Based on the salient themes extracted from the needs assessment interviews, we proposed our preliminary system design guidelines.

To motivate patients to self-report, the design of the user interface has to minimize users’ efforts and incorporate the data that allow positive interpretations (i.e., not reporting how much patients have lost). Given that patients’ lifestyles may be disrupted, automatic sensing devices (e.g., wristbands) can be added to monitor patients’ lifestyles. Moreover, it is important to incorporate visualizations for patients to correlate the impact of the gambling behaviors on their lifestyles and self-reported gambling conditions.

Given the poor communication among patients and family members when discussing gambling problems, patients would only resort to family members when they can no longer pay off their debt. To avoid feelings of shame when expressing their gambling status, one feasible direction is to leverage the phone as a buffer to progressively share individual pieces of gambling information with family members, instead of helplessly confessing the whole story all at once. This can lower the resistance and increase the willingness of patients to share the gambling data with their family members.

Finally, based on results, family members seemed unfamiliar with gambling addiction in general. Without proper understanding and knowledge of gambling addiction, family members might have incorrect reasoning or have little motivation to check the collected data, which may, in turn, result in unnecessary conflicts or misunderstandings. Therefore, inputs and knowledge from treatment experts would be critical for promoting a supportive way to utilize data and should be incorporated in the user interface.

4 CONCLUSION & FUTURE WORK

In the future, we will interview more patients and family members to refine the design of this system. We will also prototype our system and further validate its potential and feasibility through conducting a field study involving both gambling-addicted patients and their family members.

ACKNOWLEDGMENTS

This research was supported by the Ministry of Science and Technology of Taiwan (MOST 109-2221-E-007-114, 110-2221-E-007-098, and 109-2221-E-003-010-MY3).

REFERENCES

- [1] Gambling Recovery Apps. <https://www.problemgambling.ie/problem-gambling-recovery-apps.html>
- [2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101.
- [3] George Drosatos, Emily Arden-Close, Elvira Bolat, and Raian Ali. 2020. Gambling data and modalities of interaction for responsible online gambling: A qualitative study. *Journal of Gambling Issues* 44, Spring (2020), 139–169.
- [4] Gayl Humphrey, Joanna Chu, Nicki Dowling, Simone Rodda, Stephanie Merkouris, Varsha Parag, David Newcombe, Elsie Ho, Vili Nosa, Rebecca Ruwhui-Collins, et al. 2020. Manaaki—a cognitive behavioral therapy mobile health app to support people experiencing gambling problems: a randomized control trial protocol. *BMC public health* 20, 1 (2020), 191.
- [5] Sally Monaghan. 2009. Responsible gambling strategies for Internet gambling: The theoretical and empirical base of using pop-up messages to encourage self-awareness. *Computers in Human Behavior* 25, 1 (2009), 202–207.